

Statistical modeling

Module 04

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Regression: statistical assumptions

Remember the model assumptions:

- ▶ $y_i = b_0 + b_1x_{1i} + \cdots + b_px_{pi} + e_i$
- ▶ Model errors e_i are statistically independent,
- ▶ The distribution of errors e_i is $N(0, \sigma^2)$

We write

$$e_i \stackrel{i.i.d.}{\sim} N(0, \sigma^2)$$

"i.i.d." stands for "independent and identically distributed".

Regression: statistical assumptions

In particular,

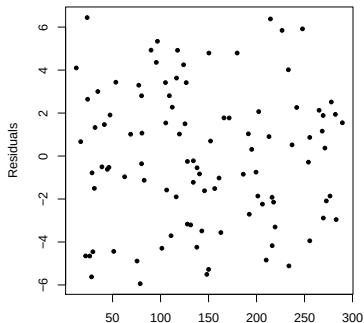
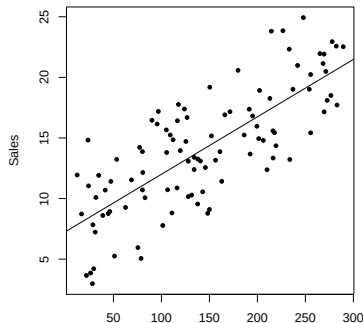
$$E(e_i) = 0, \quad \text{Var}(e_i) = \sigma^2, \quad \text{Cov}(e_i, e_{i'}) = 0, \quad i \neq i'.$$

Consequently,

1. Responses y_i are statistically independent,
2. $y_i \sim N(b_0 + b_1x_{1i} + \cdots + b_px_{pi}, \sigma^2)$.

Regression: statistical assumptions

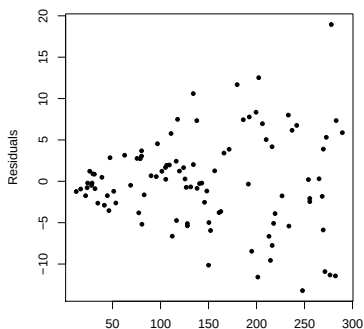
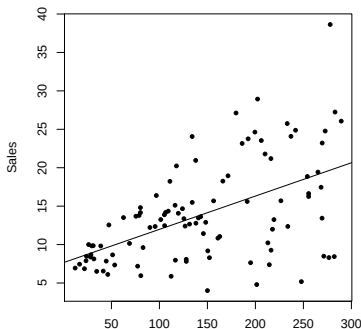
Example: in a company, product sales and TV budgets (local adv. campaign) are studied.



The residuals are normally distributed, or equivalently the observations around the curve.

Regression: statistical assumptions

Example: In that case, the residuals are normally distributed but not with a constant standard deviation. The linear model assumptions are not satisfied.



More later about how to check if these assumptions are satisfied.

The standard deviation σ

We just saw a new very important element of the regression: the standard deviation σ .

Since it is the standard deviation of the model residual, it is

- ▶ A measure of the uncertainty that we have on the response y given x (error on the prediction);
- ▶ A measure of the variability of y that is not explained by the variables in x (residual variance).

A large σ induces both:

- ▶ A large uncertainty of the prediction of y
- ▶ A large uncertainty on the estimation of β .

Its estimation of thus of central importance.

Unbiased estimation of σ^2 : s^2

Consider the sum of squares for the error:

$$SS_e = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

It can be shown that, under the statistical assumptions,

$$E(SS_e) = (n - p - 1)\sigma^2.$$

Thus,

$$s^2 = \frac{1}{n - p - 1} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

is an unbiased estimate of σ^2 :

$$E(s^2) = \sigma^2.$$

Note: $SS_e/(n - p - 1)$ is also called MS_e , Mean Squares (see later).

Unbiased estimation of σ^2 : s^2

Example: Real estate data predicting pricing: $n = 319$ (# products in the database)

$$SS_e = 2.34e^5, MS_e = s^2 = 2.34e^5/313 = 7.48e^2, s = 27.3.$$

Example: With the TV-budget data, (first case) we have

```
> summary(mod)
[...]
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	7.265872	0.656322	11.07	<2e-16 ***
TVbudg	0.047384	0.003975	11.92	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.177 on 98 degrees of freedom
Multiple R-squared: 0.5919, Adjusted R-squared: 0.5877
F-statistic: 142.1 on 1 and 98 DF, p-value: < 2.2e-16

That is, $s = 3.177$.

Prediction interval

In a situation where we want to predict y given x , instead of getting a point prediction (i.e. a single number) it is much more useful to get an interval of possible values with a certain level of probability.

Under the model assumptions, for a new x , the corresponding unknown y is

$$y \sim N(b_0 + x_1 b_1 + \cdots + c_p b_p, \sigma^2).$$

Thus, for any fixed level α ,

$$P\left\{z_{\alpha/2} \leq \frac{y - (b_0 + x_1 b_1 + \cdots + c_p b_p)}{\sigma} \leq z_{1-\alpha/2}\right\} = 1 - \alpha,$$

Prediction interval

Hence, the prediction interval at level $1 - \alpha$ of y at x is

$$b_0 + x_1 b_1 + \cdots + c_p b_p + z_{\alpha/2} \sigma \leq y \leq b_0 + x_1 b_1 + \cdots + c_p b_p + z_{1-\alpha/2} \sigma.$$

For a more compact formula, we can use the matrix notation:

$$x^T b + z_{\alpha/2} \sigma \leq y \leq x^T b + z_{1-\alpha/2} \sigma,$$

with $x = (1, x_1, \dots, x_p)^T$ and $b = (b_0, b_1, \dots, b_p)^T$.

Estimation of the prediction interval

The previous formula uses the population coefficient b and σ . In practice, replace these unknown quantities by their estimates (plug-in method):

$$x^T \hat{b} + z_{\alpha/2} s \leq y \leq x^T \hat{b} + z_{1-\alpha/2} s.$$

The most used level α is 5%, which provides a prediction interval at 95%.

Estimation of the prediction interval

Example: with the real estate case, for the following product (i.e. a new x),

$$x_1 = 2100 \text{ ft}^2$$

$$x_2 = 15 \text{ yrs}$$

$$x_3 = 4 \text{ bedrooms}$$

$$x_4 = 3 \text{ bathrooms}$$

$$x_5 = 2 \text{ cars in the garage}$$

The predicted price is $\hat{y} = 179$ ($\times 1000\$$).

The estimated prediction interval at 95% is $[125.5, 232.5]$.

Interpretation: in the population, among all the products with characteristics x , 95% have a price in this interval.

Limitations

A prediction interval does not take the estimation error into account. Indeed,

- ▶ $\hat{b}$ is not exactly equal to the true population coefficient b ;
- ▶ s^2 is not exactly equal to the true population variance σ^2 .

Reminder: common values for α and z are

α	$z_{1-\alpha/2}$
1%	2.58
5%	1.96
10%	1.64
20%	1.28

$$z_{\alpha/2} = -z_{1-\alpha/2}$$

Model building: prediction in practice

Given a data base, how can we build a good model? Roughly speaking, two steps:

1. Which independent variables x ?

- ▶ Does it make sense? (e.g. use the price of the butter in Bangladesh to predict unemployment in Switzerland)
- ▶ Is it available? (e.g. not in the data base, very costly)
- ▶ Is it reliable? (e.g. bad data quality, lots of missing values)
- ▶ Is it redundant? (e.g. use weight in kg and in lb)

2. How to include/exclude the variables?

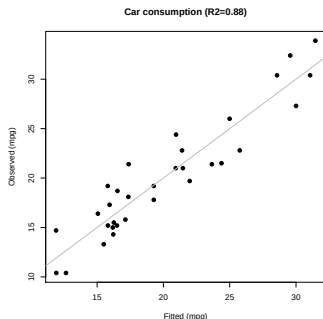
- ▶ Include the variables as they are.
- ▶ Make a change of variable (create a new one).
- ▶ Combine two variables.

Building new variables

One way to build a new variable is to make a change of variable of another one.

Example: with car data, use x_1 is weight, $x_2 = x_1^2$, x_3 is displacement, $x_4 = \log(x_3)$

$$y = 97.8 - 0.97x_1 - 0.21x_2 + 0.048x_3 - 15.8x_4 + e$$

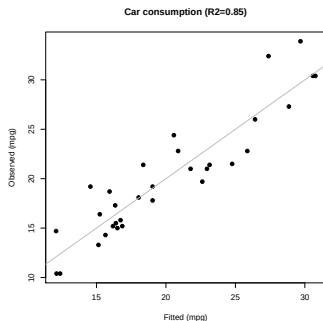


Building new variables

Another possibility is to combine two (or more) existing variables.

Example: x_1 is weight, x_2 is displacement, $x_3 = x_1x_2$,

$$y = 44.1 - 6.5x_1 - 0.06x_2 + 0.12x_3 + e$$



Building new variables

Sorm remarks:

- ▶ As long as parameters b_j are separated by a "+" or "-", it is a linear model. E.g.

$$y = 1 + x_1 + 2x_2 + 5x_2^2 + 3x_1x_2 + 8\log(x_2) + e.$$

- ▶ Linear combinations of variables (e.g. $x_4 = x_2 + 3x_3$) are redundant. E.g.

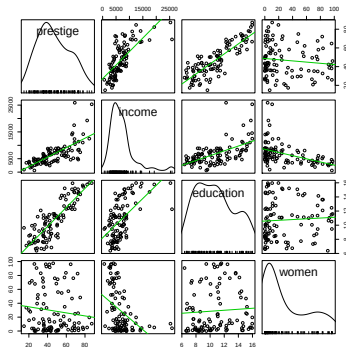
$$\begin{aligned} y &= b_0 + b_1x_1 + b_2x_2 + b_3(x_1 + x_2) + e \\ &= b_0 + (b_1 + b_3)x_1 + (b_2 + b_3)x_2 + e. \end{aligned}$$

Exploratory data analysis: scatter plots

So many models can be built... Use scatter plots to inspect potential links.

Example: find prestige relation to the other variables.

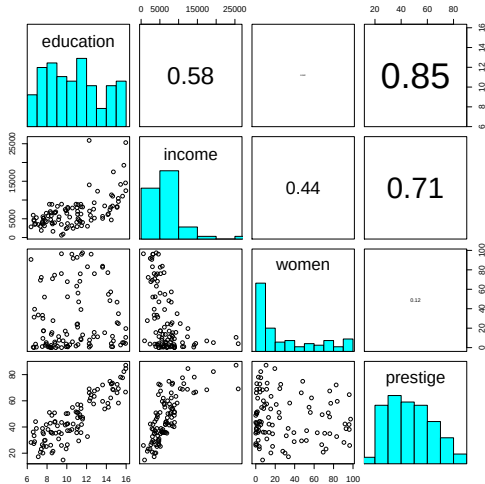
- ▶ education: avg num. of years of edu. for occupational incumbents.
- ▶ income: avg income of occupational incumbents, in \$.
- ▶ women: % of women in the occupation.
- ▶ prestige: avg prestige rating for the occupation.



source: https://rstudio-pubs-static.s3.amazonaws.com/63531_55f83f76f1104efd9b6d694f9228d55e.html

Exploratory data analysis

Use the scatter plot matrix with the correlations



Exploratory data analysis: caveats

Explanatory analysis using correlation and scatter plots are only giving a partial view on the "good" regression model.

- ▶ They are unconditional on the other variables.
E.g. in prestige data set, we see that increases with income, and that income increases with education. We cannot see if prestige would increase with prestige for a fixed level of education (or vice-versa).
- ▶ They are limited to low dimensions. E.g. what if we have 100 variables...

They are only tools for experts to handle a meaningful discussion in their domain.

Exploratory data analysis: recommendations

Recommendations:

1. Use prior expertise to define a meaningful data perimeter of the study (e.g. limit the number of variables).
2. Use explanatory analysis to try to discover how the variables are linked. Use this as guideline only.
3. Define a model and fit it. Inspect its quality (graph and R^2).
4. Review the model if needed: test alternative models¹.

In order to be able to select and use regression models, we need to develop/see some further properties.

¹See later.